COHERENCE OF THE HIGHER DIRECT IMAGES OF A COHERENT MODULE ALONG A PROPER MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Etale locality of the higher direct images and of the coherence check on the target. Properties of Spaces, label spaces-properties-lemma-pushforward-etale-base-change-modules (invoked at proof line 3850) and Same chapter spaces-cohomology.tex, label lemma-coherent-Noetherian (invoked at proof line 3852). Together they let the proof assume that the target is a Noetherian affine scheme, since the higher direct images commute with etale localization and coherence can be tested etale locally.
- The three-condition criterion for coherence of the higher direct images over a Noetherian affine base. Same chapter spaces-cohomology.tex, label lemma-property-higher-rank-cohomological-variant (invoked at proof lines 3863 and 3868). It reduces the affine case to three properties of the class of coherent modules: two-out-of-three for short exact sequences, stability under direct sums, and the reduction to a coherent module on a closed reduced irreducible subspace.
- The weak Chow lemma, producing a proper surjective cover mapping into projective space. Same chapter spaces-cohomology.tex, label lemma-weak-chow (invoked at proof line 3919). Applied to $Z \to \Spec(A)$ it yields the diagram with $\pi : Z' \to Z$ proper surjective and an immersion $i : Z' \to \mathbf{P}^n_A$, which becomes a closed immersion by properness.
- Vanishing of the higher direct images after twisting by a relatively ample invertible sheaf. Same chapter spaces-cohomology.tex, label lemma-kill-by-twisting (invoked at proof line 3944). Applied to $\mathcal{L}$ with π and with g' it gives $d \gg 0$ for which all the higher direct images of

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$\mathcal{L}^{\otimes d}$ vanish, which is what makes $\mathcal{G} = \pi_*\mathcal{L}^{\otimes d}$ usable.
- Coherence of the pushforward of a globally generated sheaf on a closed subscheme of projective space. Cohomology of Schemes, label coherent-lemma-locally-projective-pushforward (invoked at proof line 3966). It shows that $g_*\mathcal{G} = (g')_*\mathcal{L}^{\otimes d}$ is coherent, completing the verification of condition (3).
- Sandwiching of coherent modules in short exact sequences. Same chapter spaces-cohomology.tex, labels lemma-coherent-abelian-Noetherian and lemma-coherent-Noetherian-quasi-coherent-sub-quotient (invoked at proof lines 3889 and 3890). They show that a module sandwiched between two coherent modules is coherent, which is how condition (1) is verified from the long exact sequence of higher direct images.

Source reference. Original lemma 02O4: sources/stacks-chow/tags.txt, source lines 3503–3503. Navigation pointer for \ref{coherent-lemma-locally-projective-pushforward} used inside the counted span (line 3965); the argument that consumes it is retained in the span.

Source reference. Original lemma 073H: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1522–1522. Navigation pointer for \ref{lemma-affine-vanishing-higher-direct-images} used inside the counted span (line 3910); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UB: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2094–2094. Navigation pointer for \ref{lemma-coherent-Noetherian} used inside the counted span (line 3852); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UD: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2152–2152. Navigation pointer for \ref{lemma-coherent-Noetherian-quasi-coherent-sub-quotient} used inside the counted span (line 3890); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UC: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2126–2126. Navigation pointer for \ref{lemma-coherent-abelian-Noetherian} used inside the counted span (line 3889); the argument that consumes it is retained in the span.

Source reference. Original lemma 08AQ: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3809–3809. Navigation pointer for \ref{lemma-kill-by-twisting} used inside the counted span (line 3944); the argument that consumes it is retained in the span.

Source reference. Original lemma 08AN: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2779–2779. Navigation pointer for \ref{lemma-property-higher-rank-cohomological-variant} used inside the counted span (line 3863); the argument that consumes it is retained in the span.

Source reference. Original lemma 089J: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3498–3498. Navigation pointer for \ref{lemma-weak-chow} used inside the counted span (line 3919); the argument that consumes it is retained in the span.

Source reference. Original lemma 0734: sources/stacks-chow/tags.txt, source lines 9048–9048. Navigation pointer for \ref{sites-cohomology-lemma-relative-Leray} used inside the counted span (line 3908); the argument that consumes it is retained in the span.

Source reference. Original lemma 07U2: sources/stacks-chow/tags.txt, source lines 9957–9957. Navigation pointer for \ref{spaces-morphisms-lemma-closed-immersion-affine} used inside the counted span (line 3913); the argument that consumes it is retained in the span.

Source reference. Original lemma 04XY: sources/stacks-chow/tags.txt, source lines 6411–6411. Navigation pointer for \ref{spaces-morphisms-lemma-composition-proper} used inside the counted span (line 3929); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZL: sources/stacks-chow/tags.txt, source lines 6469–6469. Navigation pointer for \ref{spaces-morphisms-lemma-finite-presentation-noetherian} used inside the counted span (line 3862); the argument that consumes it is retained in the span.

Source reference. Original lemma 04CD: sources/stacks-chow/tags.txt, source lines 5687–5687. Navigation pointer for \ref{spaces-morphisms-lemma-immersion-when-closed} used inside the counted span (line 3933); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZK: sources/stacks-chow/tags.txt, source lines 6468–6468. Navigation pointer for \ref{spaces-morphisms-lemma-locally-finite-type-locally-noetherian} used inside the counted span (line 3846); the argument that consumes it is retained in the span.

Source reference. Original lemma 06G4: sources/stacks-chow/tags.txt, source lines 8273–8273. Navigation pointer for \ref{spaces-morphisms-lemma-noetherian-finite-type-finite-presentation} used inside the counted span (line 3859); the argument that consumes it is retained in the span.

Source reference. Original lemma 03KO: sources/stacks-chow/tags.txt, source lines 4684–4684. Navigation pointer for \ref{spaces-morphisms-lemma-semi-diagonal} used inside the counted span (line 3937); the argument that consumes it is retained in the span.

Source reference. Original lemma 07TZ: sources/stacks-chow/tags.txt, source lines 9954–9954. Navigation pointer for \ref{spaces-morphisms-lemma-support-finite-type} used inside the counted span (line 4006); the argument that consumes it is retained in the span.

Source reference. Original lemma 04NX: sources/stacks-chow/tags.txt, source lines 6095–6095. Navigation pointer for \ref{spaces-morphisms-lemma-universally-closed-permanence} used inside the counted span (line 3932); the argument that consumes it is retained in the span.

Source reference. Original lemma 03LX: sources/stacks-chow/tags.txt, source lines 4729–4729. Navigation pointer for \ref{spaces-properties-lemma-pushforward-etale-base-change-modules} used inside the counted span (line 3850); the argument that consumes it is retained in the span.

Source reference. Original lemma 06NJ: sources/stacks-chow/tags.txt, source lines 8538–8538. Navigation pointer for \ref{spaces-properties-lemma-quasi-separated-sober} used inside the counted span (line 3981); the argument that consumes it is retained in the span.

Source reference. Original lemma 05VQ: sources/stacks-chow/tags.txt, source lines 7559–7559. Navigation pointer for \ref{spaces-properties-lemma-stalk-pullback-quasi-coherent} used inside the counted span (line 3995); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of algebraic spaces over S with Y locally Noetherian. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Then $R^i f_* \mathcal{F}$ is a coherent $\mathcal{O}_Y$-module for all $i \geq 0$.*

Proof. We first remark that X is a locally Noetherian algebraic space by Morphisms of Spaces, Lemma 04ZK. Hence the statement of the lemma makes sense. Moreover, computing $R^i f_* \mathcal{F}$ commutes with étale localization on Y (Properties of Spaces, Lemma 03LX) and checking whether $R^i f_* \mathcal{F}$ coherent can be done étale locally on Y (Lemma 07UB). Hence we may assume that $Y = \mathrm{Spec}(A)$ is a Noetherian affine scheme.

Assume $Y = \mathrm{Spec}(A)$ is an affine scheme. Note that f is locally of finite presentation (Morphisms of Spaces, Lemma 06G4). Thus it is of finite presentation, hence X is Noetherian (Morphisms of Spaces, Lemma 04ZL). Thus Lemma 08AN applies to the category of coherent modules of X. For a coherent sheaf $\mathcal{F}$ on X we say $\mathcal{P}$ holds if and only if $R^i f_* \mathcal{F}$ is a coherent module on $\mathrm{Spec}(A)$. We will show that conditions (1), (2), and (3) of Lemma 08AN hold for this property thereby finishing the proof of the lemma.

Verification of condition (1). Let

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$$

be a short exact sequence of coherent sheaves on X. Consider the long exact sequence of higher direct images

$$R^{p-1} f_* \mathcal{F}_3 \to R^p f_* \mathcal{F}_1 \to R^p f_* \mathcal{F}_2 \to R^p f_* \mathcal{F}_3 \to R^{p+1} f_* \mathcal{F}_1$$

Then it is clear that if 2-out-of-3 of the sheaves $\mathcal{F}_i$ have property $\mathcal{P}$, then the higher direct images of the third are sandwiched in this exact complex between two coherent sheaves. Hence these higher direct images are also coherent by Lemmas 07UC and 07UD. Hence property $\mathcal{P}$ holds for the third as well.

Verification of condition (2). This follows immediately from the fact that $R^i f_*(\mathcal{F}_1 \oplus \mathcal{F}_2) = R^i f_* \mathcal{F}_1 \oplus R^i f_* \mathcal{F}_2$ and that a summand of a coherent module is coherent (see lemmas cited above).

Verification of condition (3). Let $i : Z \to X$ be a closed immersion with Z reduced and $|Z|$ irreducible. Set $g = f \circ i : Z \to \mathrm{Spec}(A)$. Let $\mathcal{G}$ be a coherent module on Z whose scheme theoretic support is equal to Z such that $R^p g_* \mathcal{G}$ is coherent for all p. Then $\mathcal{F} = i_* \mathcal{G}$ is a coherent module on X whose scheme theoretic support is Z such that $R^p f_* \mathcal{F} = R^p g_* \mathcal{G}$. To see this use the Leray spectral sequence (Cohomology on Sites, Lemma 0734) and the fact that $R^q i_* \mathcal{G} = 0$ for $q > 0$ by Lemma 073H and the fact that a closed immersion is affine. (Morphisms of Spaces, Lemma 07U2). Thus we reduce to finding a coherent sheaf $\mathcal{G}$ on Z with support equal to Z such that $R^p g_* \mathcal{G}$ is coherent for all p.

We apply Lemma 089J to the morphism $Z \to \mathrm{Spec}(A)$. Thus we get a diagram

$$\begin{array}{ccccc} Z & \xleftarrow{\pi} & Z' & \xrightarrow{i} & \mathbf{P}^n_A \\ & {\scriptstyle g}\searrow & \downarrow{\scriptstyle g'} & \swarrow & \\ & & \mathrm{Spec}(A) & & \end{array}$$

with $\pi : Z' \to Z$ proper surjective and i an immersion. Since $Z \to \mathrm{Spec}(A)$ is proper we conclude that g' is proper (Morphisms of Spaces, Lemma 04XY). Hence i is a closed immersion (Morphisms of Spaces, Lemmas 04NX and 04CD). It follows that

the morphism $i' = (i, \pi) : \mathbf{P}^n_A \times_{\mathrm{Spec}(A)} Z' = \mathbf{P}^n_Z$ is a closed immersion (Morphisms of Spaces, Lemma 03KO). Set

$$\mathcal{L} = i^*\mathcal{O}_{\mathbf{P}^n_A}(1) = (i')^*\mathcal{O}_{\mathbf{P}^n_Z}(1)$$

We may apply Lemma 08AQ to $\mathcal{L}$ and π as well as $\mathcal{L}$ and g'. Hence for all $d \gg 0$ we have $R^p\pi_*\mathcal{L}^{\otimes d} = 0$ for all $p > 0$ and $R^p(g')_*\mathcal{L}^{\otimes d} = 0$ for all $p > 0$. Set $\mathcal{G} = \pi_*\mathcal{L}^{\otimes d}$. By the Leray spectral sequence (Cohomology on Sites, Lemma 0734) we have

$$E_2^{p,q} = R^p g_* R^q \pi_* \mathcal{L}^{\otimes d} \Rightarrow R^{p+q}(g')_*\mathcal{L}^{\otimes d}$$

and by choice of d the only nonzero terms in $E_2^{p,q}$ are those with $q = 0$ and the only nonzero terms of $R^{p+q}(g')_*\mathcal{L}^{\otimes d}$ are those with $p = q = 0$. This implies that $R^p g_*\mathcal{G} = 0$ for $p > 0$ and that $g_*\mathcal{G} = (g')_*\mathcal{L}^{\otimes d}$. Applying Cohomology of Schemes, Lemma 02O4 we see that $g_*\mathcal{G} = (g')_*\mathcal{L}^{\otimes d}$ is coherent.

We still have to check that the support of $\mathcal{G}$ is Z. This follows from the fact that $\mathcal{L}^{\otimes d}$ has lots of global sections. We spell it out here. Note that $\mathcal{L}^{\otimes d}$ is globally generated for all $d \geq 0$ because the same is true for $\mathcal{O}_{\mathbf{P}^n}(d)$. Pick a point $z \in Z'$ mapping to the generic point ξ of Z which we can do as π is surjective. (Observe that Z does indeed have a generic point as $|Z|$ is irreducible and Z is Noetherian, hence quasi-separated, hence $|Z|$ is a sober topological space by Properties of Spaces, Lemma 06NJ.) Pick $s \in \Gamma(Z', \mathcal{L}^{\otimes d})$ which does not vanish at z. Since $\Gamma(Z, \mathcal{G}) = \Gamma(Z', \mathcal{L}^{\otimes d})$ we may think of s as a global section of $\mathcal{G}$. Choose a geometric point $\overline{z}$ of Z' lying over z and denote $\overline{\xi} = g' \circ \overline{z}$ the corresponding geometric point of Z. The adjunction map

$$(g')^*\mathcal{G} = (g')^* g'_* \mathcal{L}^{\otimes d} \longrightarrow \mathcal{L}^{\otimes d}$$

induces a map of stalks $\mathcal{G}_{\overline{\xi}} \to \mathcal{L}_{\overline{z}}$, see Properties of Spaces, Lemma 05VQ. Moreover the adjunction map sends the pullback of s (viewed as a section of $\mathcal{G}$) to s (viewed as a section of $\mathcal{L}^{\otimes d}$). Thus the image of s in the vector space which is the source of the arrow

$$\mathcal{G}_{\overline{\xi}} \otimes \kappa(\overline{\xi}) \longrightarrow \mathcal{L}^{\otimes d}_{\overline{z}} \otimes \kappa(\overline{z})$$

isn't zero since by choice of s the image in the target of the arrow is nonzero. Hence ξ is in the support of $\mathcal{G}$ (Morphisms of Spaces, Lemma 07TZ). Since $|Z|$ is irreducible and Z is reduced we conclude that the scheme theoretic support of $\mathcal{G}$ is all of Z as desired. □